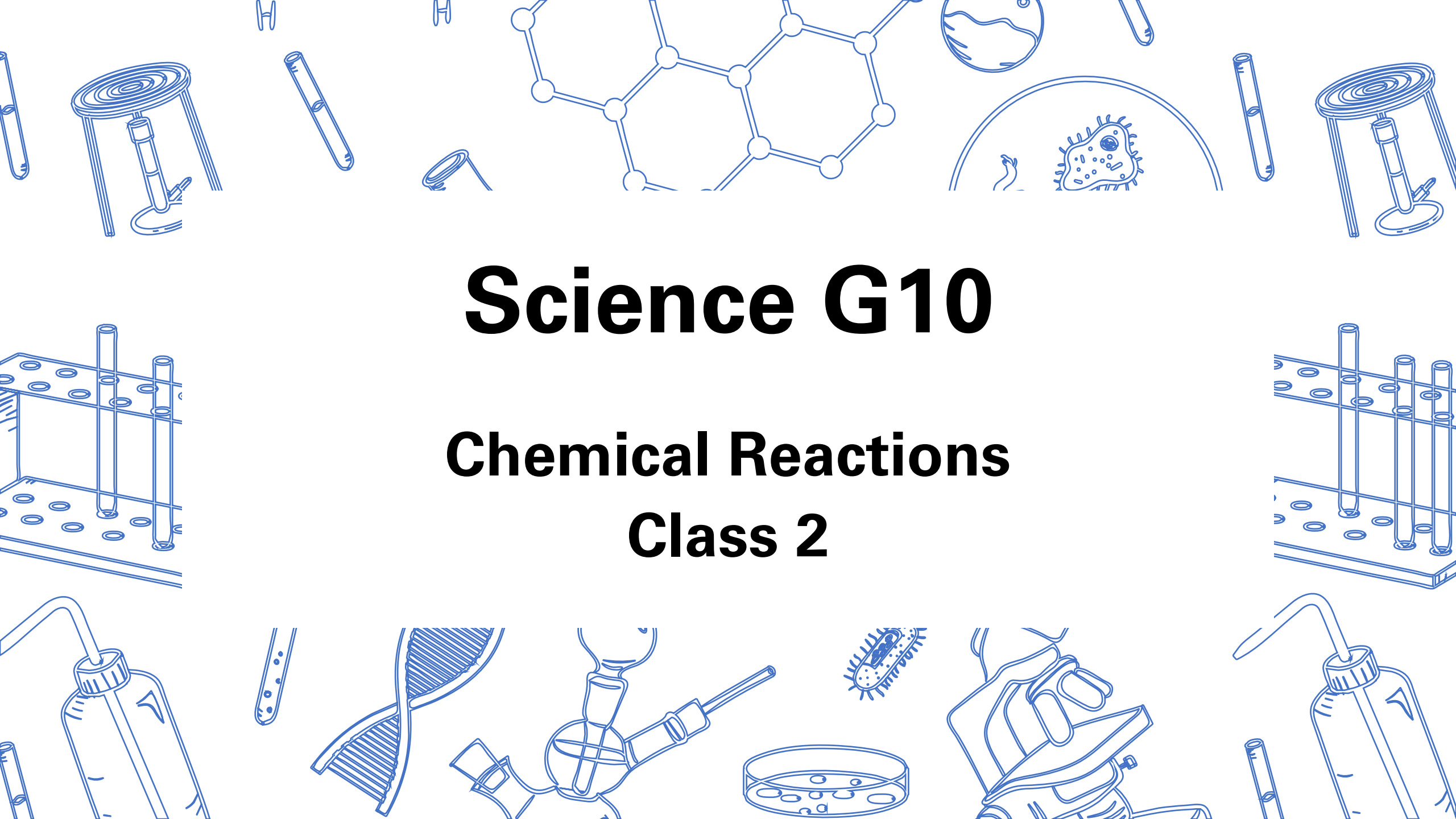


The background is a light blue collage of various scientific illustrations. At the top center is a molecular structure with interconnected hexagons. To its right is a petri dish containing a microorganism. Below the molecular structure is a DNA double helix. To the right of the DNA is a hand holding a microscope. At the bottom center is a petri dish with several small circles representing cells. To the left of the DNA is a test tube with bubbles. To the right of the hand holding the microscope is another petri dish with cells. In the top left and right corners are test tubes and a Bunsen burner. In the bottom left and right corners are test tubes and a flask with a tube. The central text is in a bold, black, sans-serif font.

Science G10

Chemical Reactions

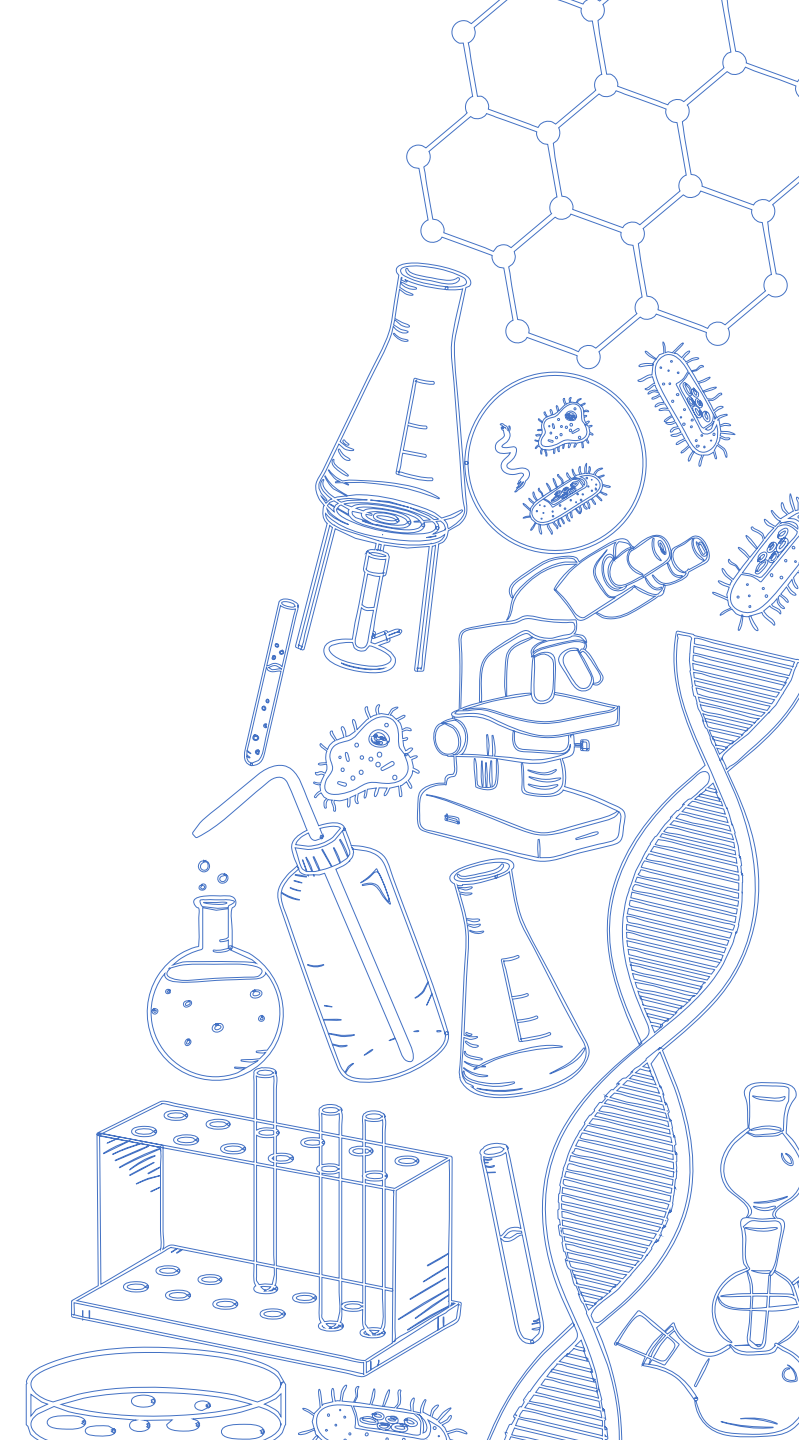
Class 2

Agenda

1. Quiz 2

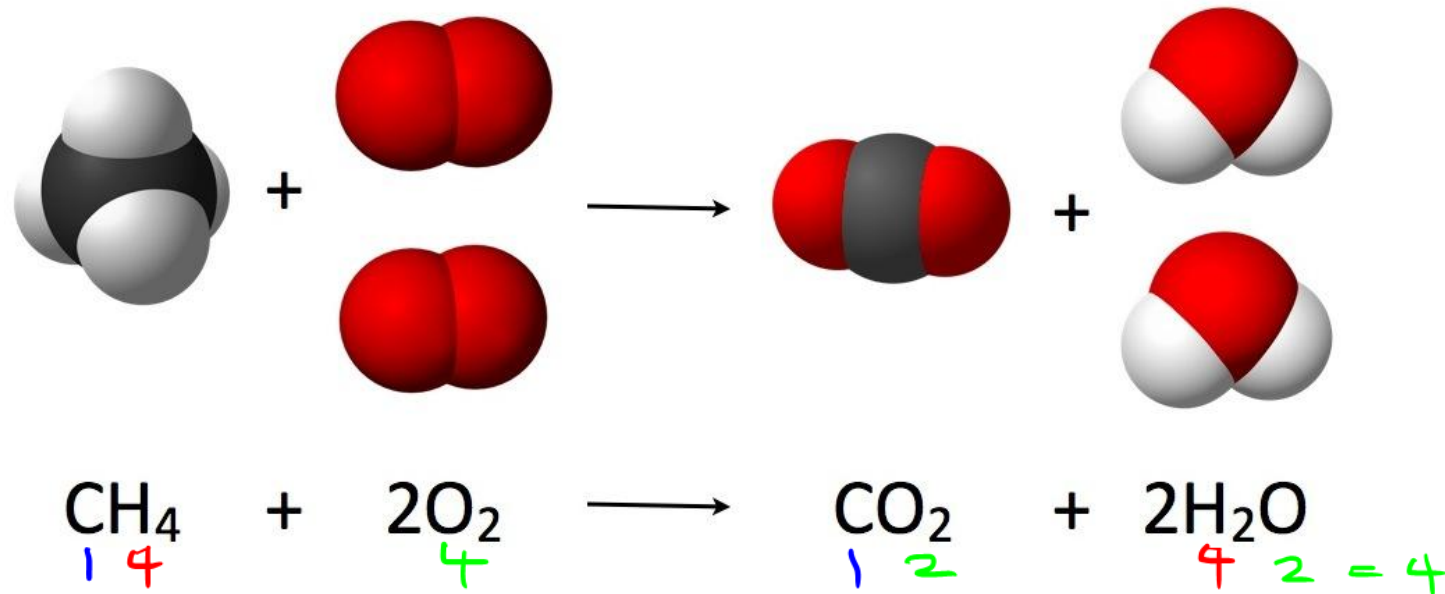
2. Lesson 2

- Law of Conservation of Mass
- Balancing Chemical Reactions
- Types of Chemical Reactions
 - Synthesis
 - Decomposition
 - Combustion
 - Single Displacement
 - Double Displacement

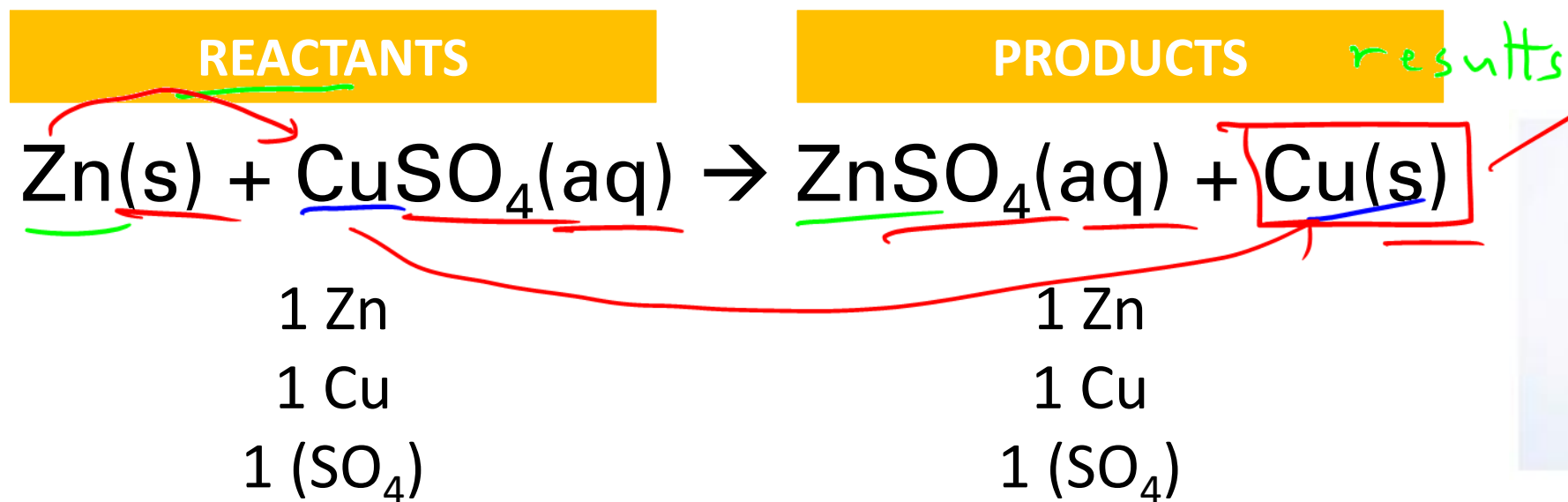


Law of Conservation of Mass

- In any given reaction, the total mass of the reactants equals the total mass of the products
- Atoms cannot be created or destroyed



Chemical Reactions



Handwritten notes:

HCl	hydrogen chloride
HCl(aq)	hydrochloric acid

Single displacement reaction

State Symbols

important when dealing
with acids

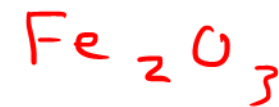
- State symbols are often written behind the chemical formula to indicate the state of the substance



State Symbol	Meaning
(s)	Solid
(l)	Liquid/Molten
(g)	Gaseous
(aq)	Aqueous (dissolved in water)

NaCl(aq)

invisible



Balancing Chemical Reactions

Word Equation	Iron reacts with oxygen to form <u>iron(III) oxide</u>				
Skeleton Equation	$4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$ <i>↓ 1 atomic</i> <i>↑ Fe³⁺ O²⁻</i> <i>multivalent ionic compound</i>				
<i>skip</i> Number of Atoms	<table border="1"><tr><td>1 Fe</td><td>2 Fe</td></tr><tr><td>2 O</td><td>3 O</td></tr></table>	1 Fe	2 Fe	2 O	3 O
1 Fe	2 Fe				
2 O	3 O				
Balanced Reaction (add <u>coefficients</u>)	$4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$ <i>2 1.5 1</i> <i>4 3 2</i> <i>× 2</i> <i>You CANNOT change the subscripts when you are balancing. Only you can add or change the coefficients.</i>				



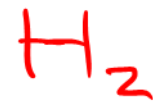
General Rules for Balancing Chemical Reactions:

- Only add coefficients, do not add subscripts ✓
- ~~Balance non-hydrogens and non-oxygens first~~
- Balance polyatomic ions as a unit ✓ H_3O_2
- All coefficients in the balanced chemical reaction need to be in simplest form as positive integers $\text{no H}_{1.5}\text{O}$
- If chemical reaction cannot be balanced, check if the crisscrossing was done correctly

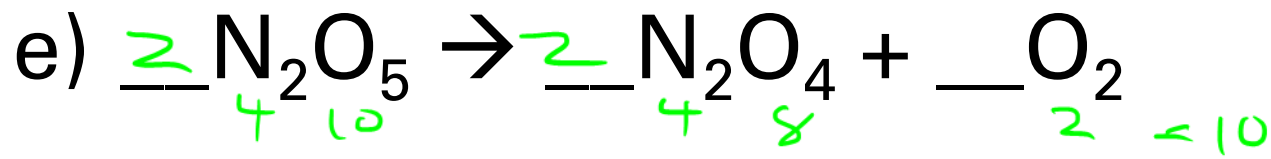
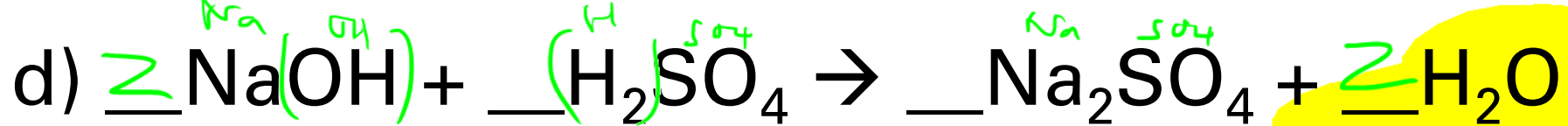
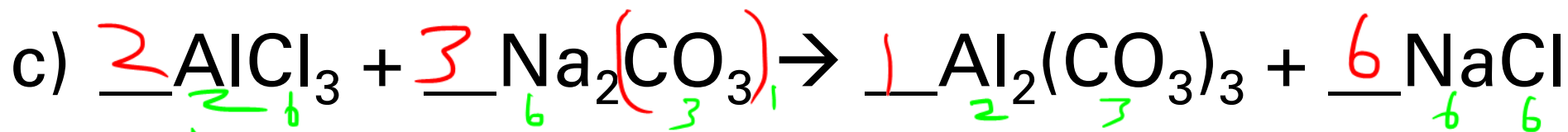
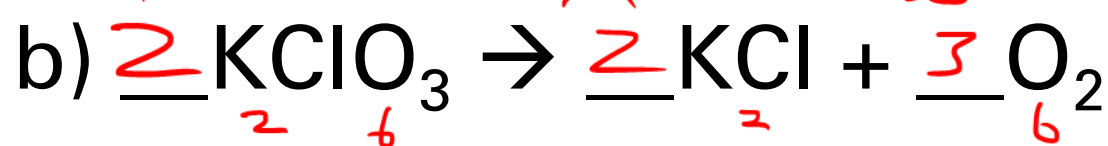
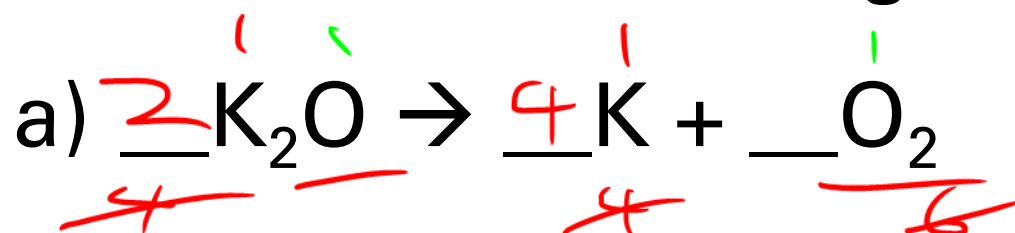
— always start with the atom with the least terms, if < 1, you pick



Checkpoint

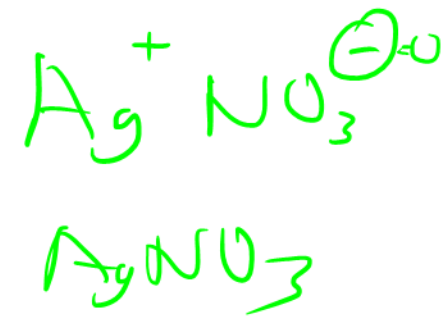


Balance the following chemical reactions:





Checkpoint



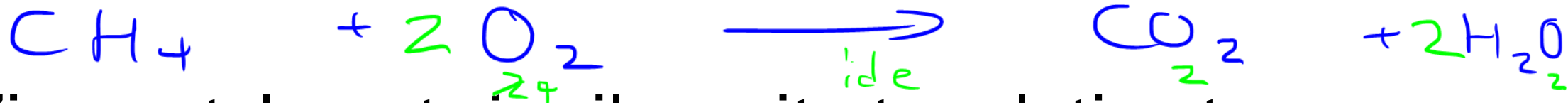
Write the balanced chemical reaction of:



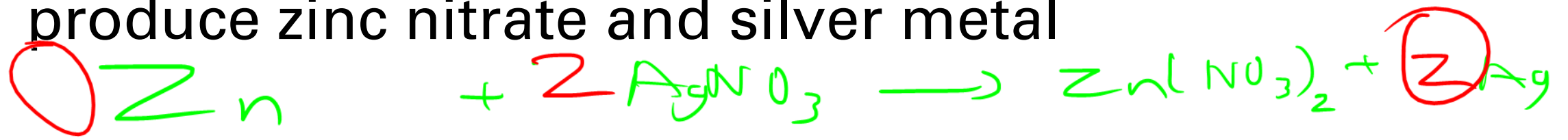
a) Magnesium with oxygen



b) Methane (CH_4) burns in oxygen to produce carbon dioxide and water



c) Zinc metal reacts in silver nitrate solution to produce zinc nitrate and silver metal



5g

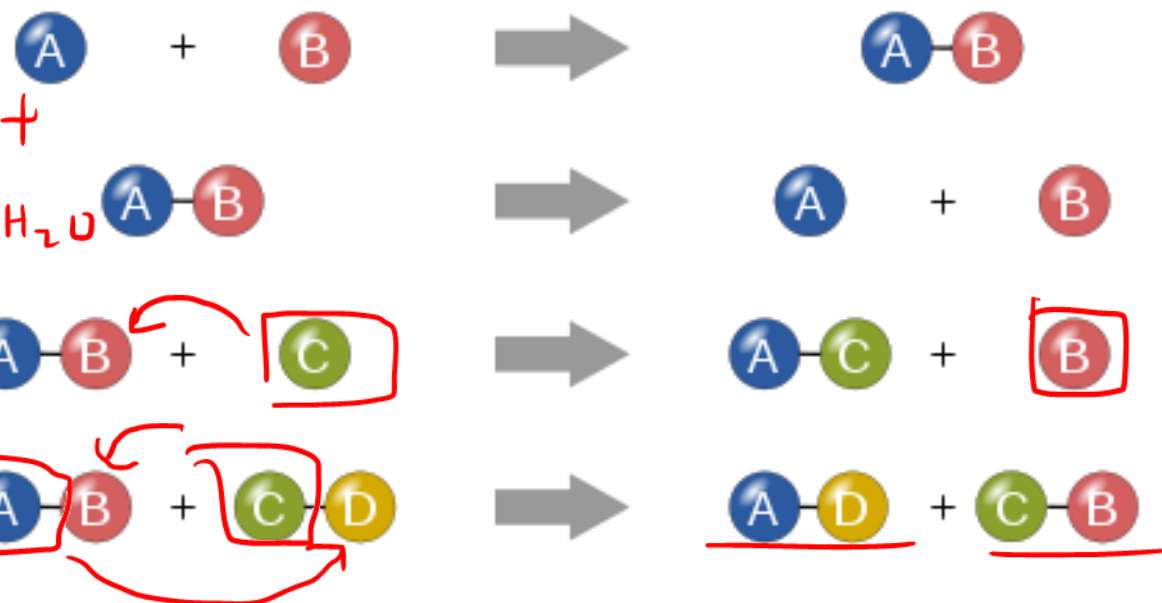
1

?g

2

Types of Chemical Reactions

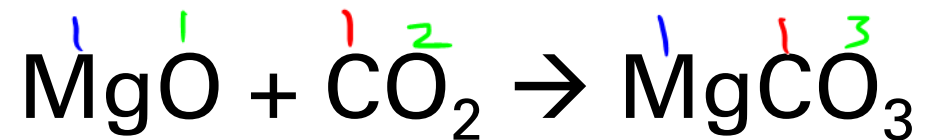
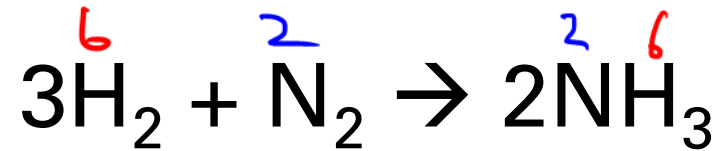
- Synthesis *combine*
- Decomposition *break apart*
- ~~X~~ • Combustion *$x + O_2 \rightarrow CO_2 + H_2O$*
- Single Displacement
- Double Displacement
 - Neutralization



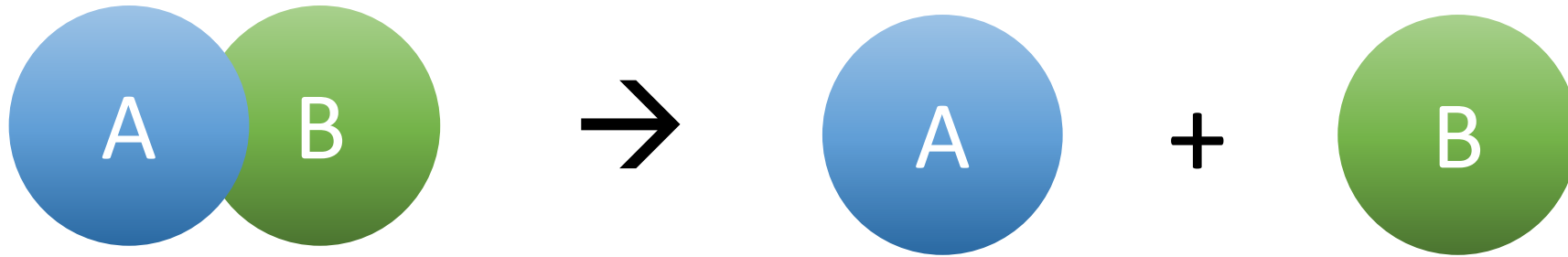
Synthesis



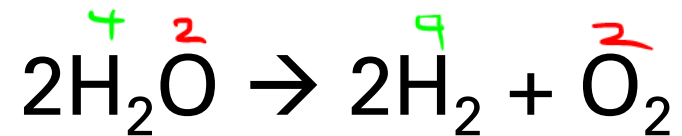
Examples:



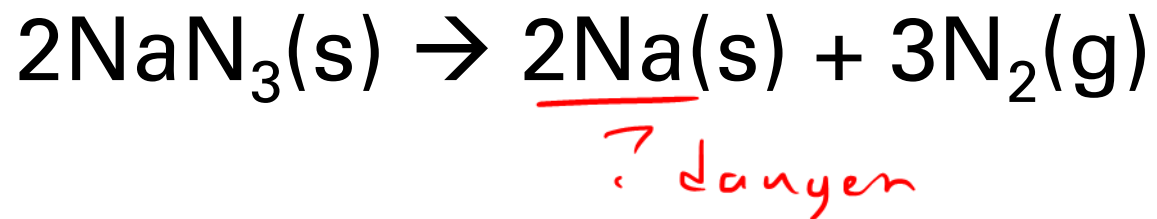
Decomposition



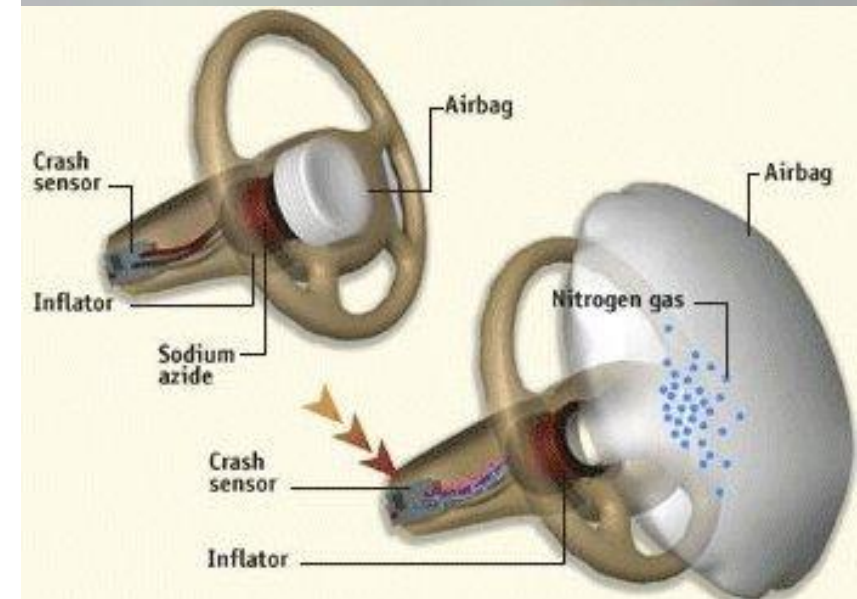
Examples:



Decomposition of Sodium Azide



- Airbags contain sodium azide, $\text{NaN}_3(\text{s})$
- During a collision, electrical energy triggers the rapid decomposition of $\text{NaN}_3(\text{s})$ into nitrogen gas to inflate the airbag





Checkpoint



Write the balanced chemical equation for the following and classify the type of reaction:

a) Zinc chloride $\rightarrow$ zinc + chlorine

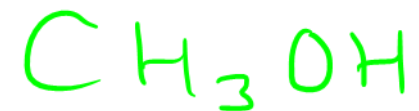


b) Potassium oxide + water $\rightarrow$ potassium hydroxide



Combustion

- **Combustion** is a reaction in which a substance reacts with oxygen gas, releasing energy in the form of light and heat
- Two types:
 - **Complete combustion** occurs when there is sufficient oxygen
 - **Incomplete combustion** occurs where there is insufficient oxygen



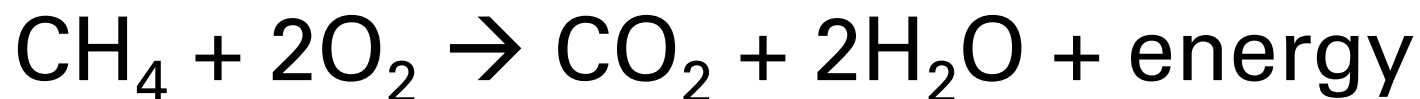


Complete Combustion of Hydrocarbons

- Hydrocarbons are compounds that contain carbons and hydrogens
- Complete combustion of hydrocarbons produces CO_2 , H_2O , and energy
- Flame is blue in colour with little smoke

Complete combustion

Methane



Air Hole Open



Checkpoint



Assuming complete combustion, write the balanced chemical reaction for the combustion of:

a) Propane (C_3H_8)

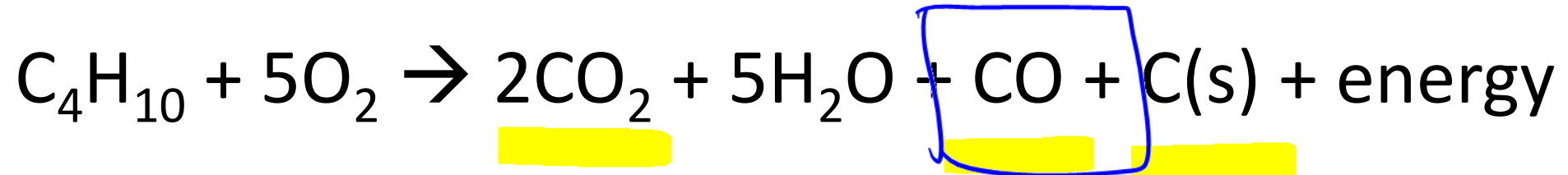


b) Octane (C_8H_{18})



Incomplete Combustion of Hydrocarbons

- Incomplete combustion of hydrocarbons produces CO_2 , CO , H_2O , C(s) , and energy
- Flames from incomplete combustion tend to be yellow or orange in colour and are smokier due to unburned carbon



carbon monoxide

Dangers of Carbon Monoxide

- Carbon monoxide displaces oxygen in the blood and deprives the heart, brain, and other vital organs of oxygen
- Carbon monoxide is colourless, odourless, and tasteless

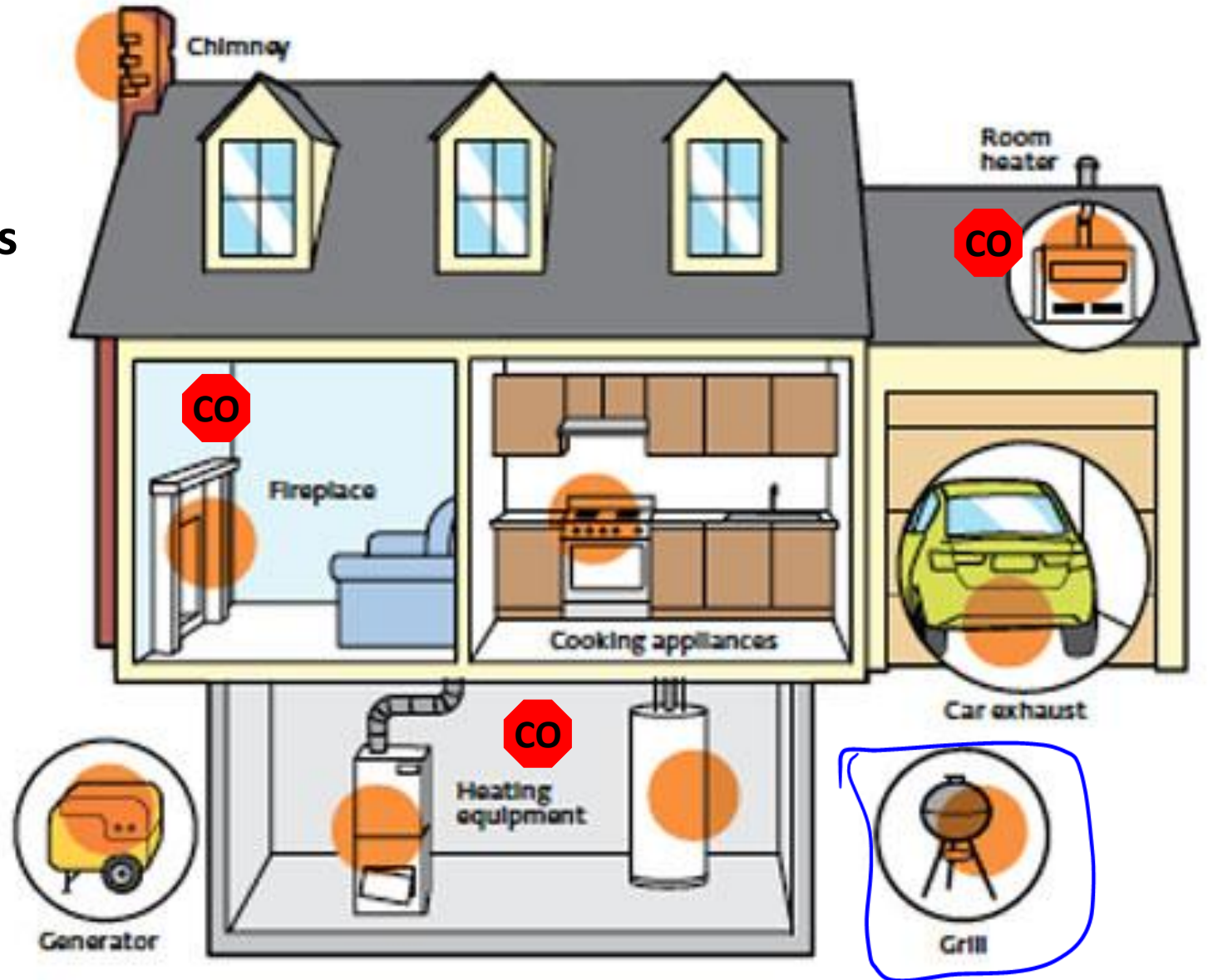
Symptoms of carbon monoxide poisoning:



▶ POTENTIAL DANGER ZONES

CO Placement of CO(g) alarms

- It is recommended that at least one alarm is installed on every level



BY D'ARCY JENISH

Among his friends and neighbors in Wilkie, Sask., 43-year-old Robert Latimer was considered a typical Prairie farmer. Hardworking, clean-living and self-reliant, he would repair his vehicles or replace the barn roof himself rather than hire someone. At the local grain elevator, where he sold the wheat grown on his 1,000-acre farm 170 km west of Saskatoon, he was known as "Laddie" and viewed more as a friend than a customer. But last week, in a courtroom in Flettford, a half-hour drive up Highway 20, another group of his peers—as 11 members jury—passed a harsh judgment on Latimer. They found him guilty of second-degree murder in the October, 1993, mercy killing of his 13-year-old daughter, Tracy, who was severely mentally and physically disabled by cerebral palsy. Their verdict, and the life sentence imposed by Justice Ross Winter—with no parole for 10 years—shocked Latimer's friends and neighbors and reignited a long-simmering debate about the ethics of euthanasia in Canadian society.

The facts in the Latimer case were never in dispute—he confessed the crime to police. But his motive—compassion for a child in constant pain—turned a potentially routine trial into an emotional, highly publicized and controversial event. It occurred, coincidentally, as a Senate committee was preparing to receive public hearings on Nov. 23 in Ottawa on euthanasia and doctor-assisted suicide. The trial attracted the interest of right-to-life organizations, which want the Criminal Code amended so that the terminally ill can legally end their lives. Advocates for the disabled, who generally oppose euthanasia, applauded the Latimer verdict as a powerful message that the lives of the disabled must be protected. "I'm sure he had his reasons," said Lloyd Sumner, 44, a volunteer with the Halton Regional Cerebral Palsy Association, who is affected with the disorder. "But when somebody plays God, it's scary."

Many Canadians, however, viewed Latimer as the victim of a law that was too rigid, and a sentence that was too harsh (page 26). He was awaiting a bail hearing, pending an appeal that could go all the way to the Supreme Court. His only other option for leniency would be to apply to Justice Minister Allan Rock, who could grant a conditional pardon on compassionate grounds under a rarely used provision of the Criminal Code. The verdict also led to suggestions, within the legal community and on open-line radio shows, that the law should be amended to give judges some discretion in sentencing people convicted of murder. "What happened in Ottawa last week, Rock said that he opposes the idea,

WHAT WOULD YOU DO?

In Saskatchewan, a wrenching verdict of murder reignites a long-simmering debate about mercy killing



Robert and Tracy Latimer: If she had started to cry, I would have taken her out.

Meanwhile, residents of Wilkie, population 1,500, were howling up against the windy weather and rallying around a local family. "I thought he might get a year or two," said Janey Sherman. Robert, before heading out into the winds and blowing snow in an old green truck, "This sentence is like using a sledgehammer on a flea." The Latimer family received dozens of calls from sympathetic people across the country, many offering donations to help cover legal expenses. "A lot of people have sent money, but Bob won't accept it," his wife, Laura, told reporters who camped on the family home. "He asked that it be sent back, but he appreciates the moral support." In an interview with *Maclean's* (page 23), she added: "I know I married the right person. It's awful to go to bed at night when he's not there. I support my husband. I know what a very loving dad he was to Tracy. This whole thing has brought us closer."

Others, particularly those who work with the disabled and re-

intervention, as long as she was. "Tracy would not have been in this situation if people had not used herose muscles all along," said Elizabeth Kluge, an ethicist at the University of Victoria and a former director of ethics and legal affairs for the Canadian Medical Association. "Someone should have stopped along the way and asked, 'Should we be doing this?' By keeping her alive to face a declining quality of life, they were committing what ethicists call the 'bargain of continuing existence.' In the end, the parents faced a situation in which they thought that they had no choice."

For the past year, while he was free on bail and awaiting trial on a murder charge, Robert Latimer ran his business and participated in the life of his community as he always had. A native of Wilkie, he grew up on the farm that his father, Wilbur, who died in October at age 67, and his mother, Mae, 92, acquired in 1948. The birth of four children, Latimer bought the family farm 18 years ago while his siblings moved away to pursue their careers.

very threatened by the possibility that he would be acquitted. Not to send him to prison would send a horrible message to disabled people that their lives are of no value, that they are meaningless to society."

Shah also pointed out that the debate over the trial has turned public attention away from another crucial issue facing the physically and mentally disabled—the inadequacy and deteriorating government funding for support programs. Some parents, like Saskatchewan mother Brenda Barreiros, who has a 10-year-old daughter with severe cerebral palsy, contend that finding good caregivers is difficult in cities like Regina, and next to impossible in rural areas. Michael Burgess, an associate professor of bioethics at the University of Calgary, agrees. "While government cutbacks, we are simply transferring the burden of costs from the public purse to private families. We should be asking why we, as a society, are failing to support the disabled and their caregivers."

But some medical experts contend that Canadians must be prepared to discuss the legal and ethical issues of euthanasia because the Latimer case is just the tip of the iceberg, and he was simply the person who got caught. They maintain that passive euthanasia, in which treatment is withheld, is widely practiced in Canadian hospitals, usually by medical professionals. "It's done all the time," said Alister Browne, an ethicist at the University of British Columbia in Vancouver. "Terminally ill patients are often off for severely defective newborns, leading tubes are disconnected, elderly patients are not resuscitated. It's part of routine medical care."

Other prominent ethicists contend that Tracy Latimer never should have been kept alive, through repeated surgery and other medical intervention, as long as she was. "Tracy would not have been in this situation if people had not used herose muscles all along," said Elizabeth Kluge, an ethicist at the University of Victoria and a former director of ethics and legal affairs for the Canadian Medical Association. "Someone should have stopped along the way and asked, 'Should we be doing this?' By keeping her alive to face a declining quality of life, they were committing what ethicists call the 'bargain of continuing existence.' In the end, the parents faced a situation in which they thought that they had no choice."

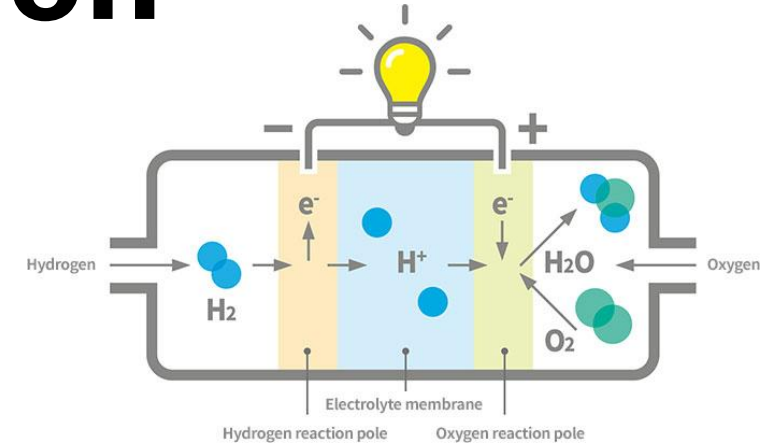
For the past year, while he was free on bail and awaiting trial on a murder charge, Robert Latimer ran his business and participated in the life of his community as he always had. A native of Wilkie, he grew up on the farm that his father, Wilbur, who died in October at age 67, and his mother, Mae, 92, acquired in 1948. The birth of four children, Latimer bought the family farm 18 years ago while his siblings moved away to pursue their careers.



- October 24, 1993, Robert Latimer killed his 13-year old daughter Tracy Latimer by piping the exhaust fumes containing CO(g) into the interior of the car where she was placed
- Tracy had a severe form of cerebral palsy and suffered considerable pain
 - Father killed her to relieve her of her pain
- Triggered debates around health ethics and euthanasia

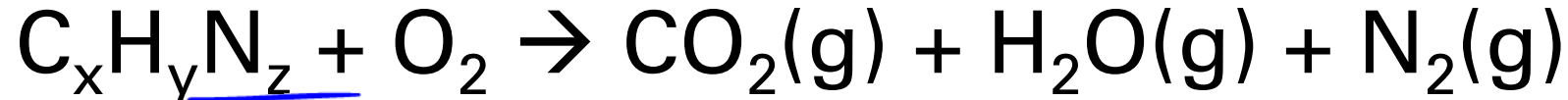
Other Forms of Combustion

Combustion of Hydrogen (Fuel Cells)



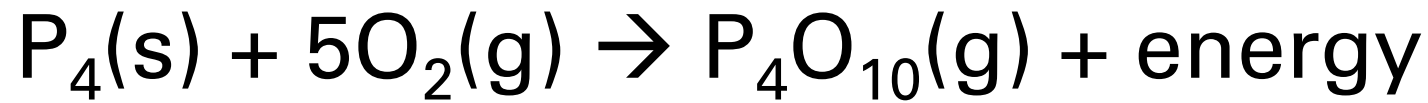
dangerous

Combustion of Nitrogen-Containing Hydrocarbons



T

Combustion of Phosphorus

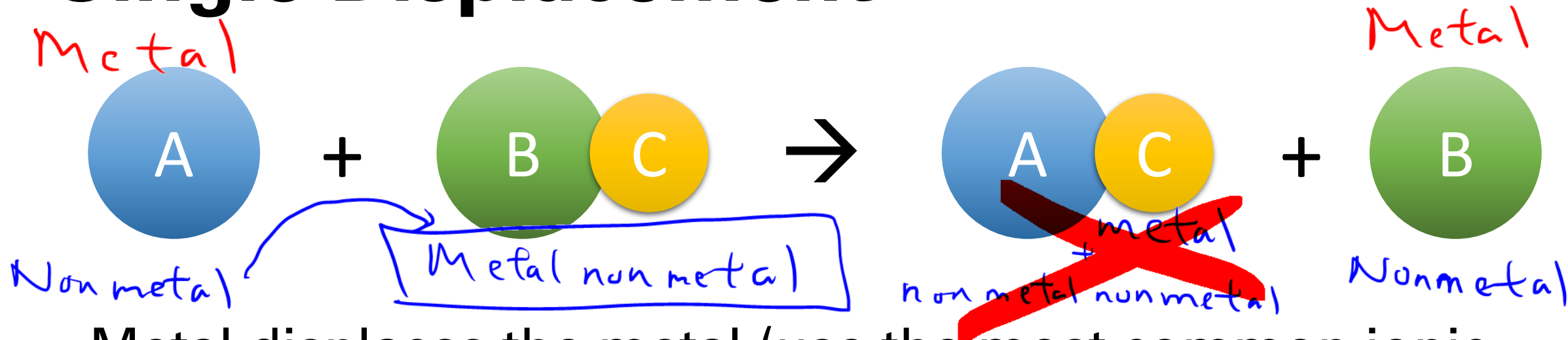


matchsticks

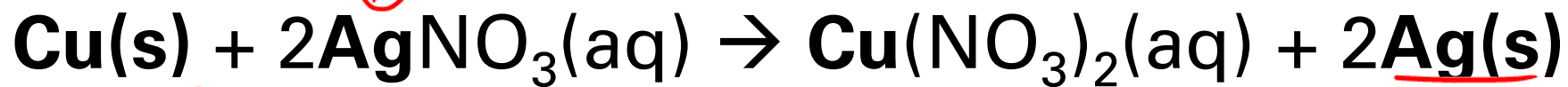


rough surface

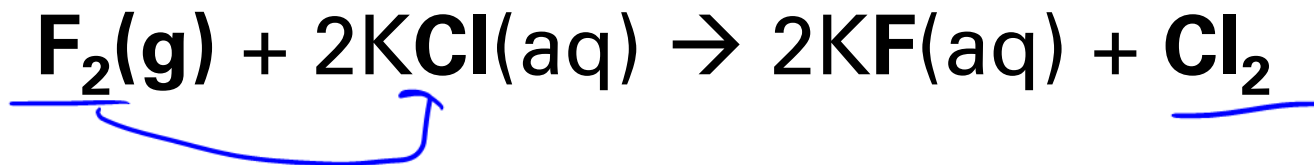
Single Displacement



- Metal displaces the metal (use the most common ionic charge for multivalent metals unless indicated otherwise)



- Nonmetal displaces the nonmetal

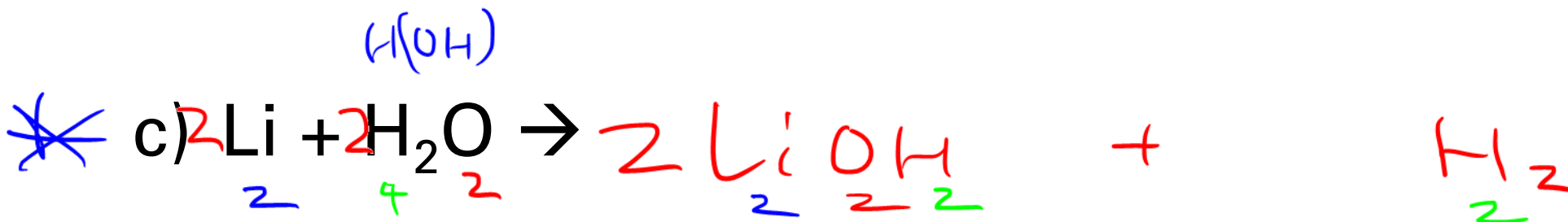




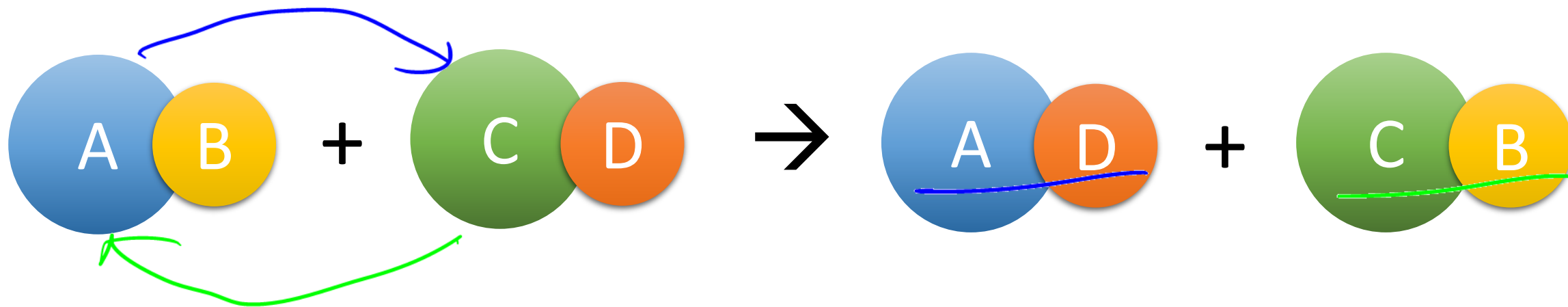
Checkpoint



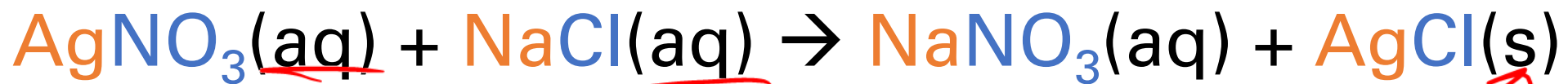
Predict the products of the following single displacement reactions and balance:



Double Displacement



- **Metals** trade places or **nonmetals** trade places

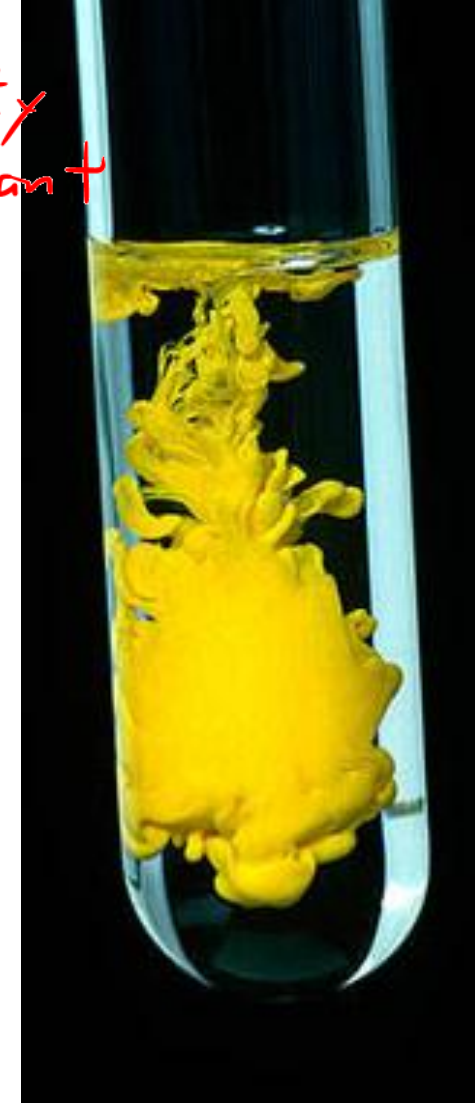
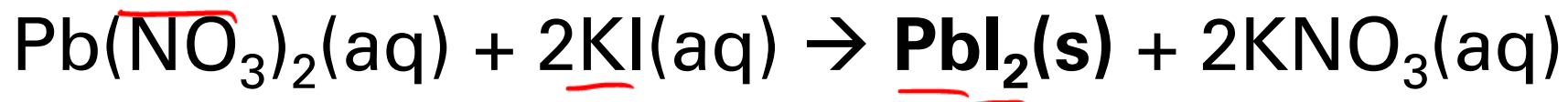


- Double displacement reactions occur if:
 - A solid precipitate forms OR H₂O forms OR gas forms

Precipitate

Gr II : K_{sp} solubility constant

- **Precipitate** is a solid formed from the reaction of two solutions; an insoluble compound



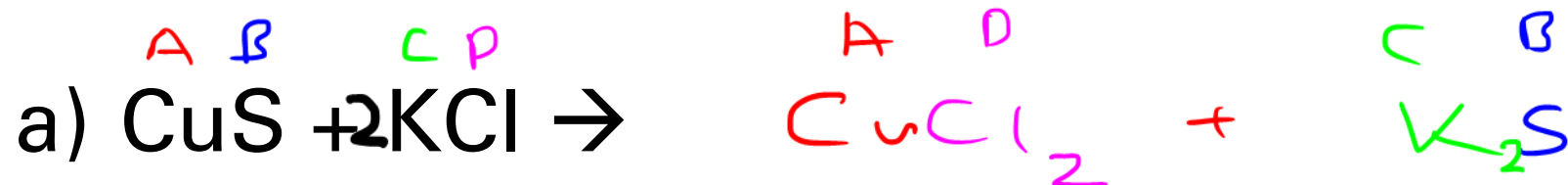
$\text{PbI}_2(\text{s})$ is a yellow precipitate



Checkpoint

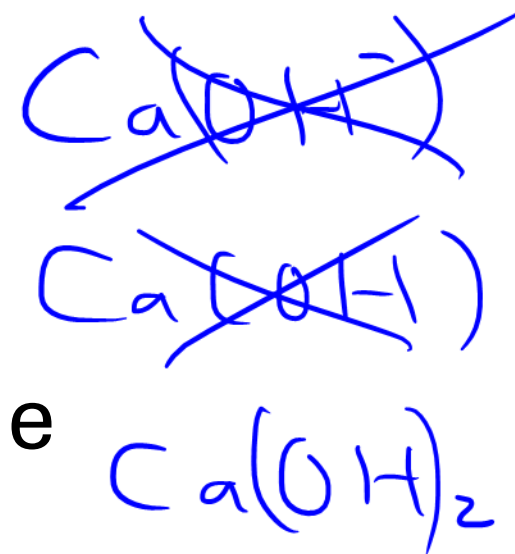


Predict the products of the following double displacement reactions and balance:





Checkpoint



Write the balanced chemical equation for the following and classify the type of reaction:

a) Silver nitrate + sodium phosphate → silver phosphate precipitate + sodium nitrate



b) Calcium + water → hydrogen gas + calcium hydroxide



What I Learned Today:

- ❑ Law of Conservation of Mass
- ❑ Balancing Chemical Reactions
- ❑ Types of Chemical Reactions
 - Synthesis
 - Decomposition
 - Combustion (Complete and Incomplete)
 - Single Displacement
 - Double Displacement

Due next class: Class 2 Homework

